Sahaj Bhadja

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Summary

Software Engineer with 2+ years building customer-facing **React** interfaces and **Python** backends with Flask, FastAPI, and Django on **AWS**, integrating AI models, writing automated tests, and collaborating through code reviews.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **Python** FastAPI services and data integration pipelines across PostgreSQL, SQL Server, and cloud storage with validation and CI/CD-backed releases, sustaining 99%+ reliability over 5K+ monthly jobs.
- Designed multi-tenant schemas and indexed queries for 500+ users, cutting **dashboard latency** from 1.2s to 350ms, and restructured async LLM job processing to triple throughput across 1K+ jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built customer-facing **React** views and a Python Flask backend on **AWS** for upload, rubric, and grading workflows, serving 150+ instructors and 10K+ daily requests with stable peak-load responses.
- Integrated **AI** models including GPT-4o, Claude, and Qwen-VL with OCR fallbacks, per-call cost tracking, and multithreaded batch processing, cutting instructor feedback turnaround 60% at 90%+ grading consistency.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned **AWS** Glue and S3 ETL workflows that consolidated user activity logs into analysis-ready formats, boosting pipeline efficiency 68% and cutting storage costs 15% for downstream APIs.
- Wrote 20+ PyTest and Mockito suites run nightly in Jenkins and refined API performance with pagination and caching, improving response times 30% while collaborating in **Agile** sprints with Microsoft stakeholders.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **Django and React** platform with RESTful APIs and real-time WebSocket updates for customer records, leads, billing, and **email and SMS notifications**, syncing 2M+ records across locations with fewer sync conflicts.
- Took part in **code review** and rollout across 6+ core workflows, gating releases on Jest and PyTest tests in Jenkins CI/CD, reaching 95%+ successful build validation and cutting deployment time 50%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built a **React and TypeScript** web dashboard and Python Flask APIs for bookings, pricing, and coupons with Firebase real-time updates, keeping core responses under 200ms across 50K+ booking records.
- Triggered customer booking **notifications** through Firebase Cloud Messaging and automated refunds through idempotent payment webhooks, cutting manual refund handling about 70% and speeding mobile booking confirmations 40%.

Skills

Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS, Java
Frontend & Tools: React, Next.js, TailwindCSS, React Native, WebSockets, Git, GitHub Actions, Agile
Backend: Django, Flask, FastAPI, REST APIs, Microservices, Node.js, Stripe, Docker
Cloud & Data: AWS (S3, RDS, Glue), PostgreSQL, MySQL, SQL Server, MongoDB, Firebase, ETL Pipelines
AI & Quality: LLM APIs, Prompt Engineering, OCR, PyTest, Jest, Selenium, Unit Testing, Code Reviews, CI/CD, Jenkins

Projects

Email Campaign Scheduler | *Python, Django, React, PostgreSQL, Celery, AWS SES*

- Designed a **Django** and React app that segments audiences, schedules email campaigns through AWS SES, handles bounce and unsubscribe webhooks, and tracks opens and clicks in campaign **analytics** dashboards.

AI Subject Line Assistant | *Python, Django REST, React, LLM APIs, PostgreSQL*

- Prototyped an **LLM**-powered tool that drafts email subject lines and preview text from a campaign brief, runs **A/B tests** on generated variants, and requires marketer approval before anything is sent.

PiXMod Live | *Next.js, Stripe, CLIP, SAM, Stable Diffusion, MongoDB*

- Built an AI-powered **SaaS** image platform with SAM segmentation, Stable Diffusion editing, CLIP search, and Stripe subscription billing behind a responsive Next.js interface, increasing premium conversion 18%.